Temp + Gravity Module — Complete Parts List

Everything needed to build one floating unit that reports **temperature and specific gravity**. Schematic: `schematic-gravity-module.html`. Firmware: `firmware/fermentation_tilt_v1/`.

> **This is the production BOM — SMD parts for a fabbed board.** To build a working float on a breadboard first, see `PROTOTYPE_TEMP_GRAVITY.md`.

Also published as a checkable order list: <https://claude.ai/code/artifact/30287fa3-41c6-4aa6-bf64-7ca5688c4331>

Costs are single-unit estimates in USD. ZAR figures use the ~25% landed markup convention from `docs/IMPLEMENTATION_PLAN.md` (shipping + customs on imported electronics), not the raw FX rate.

> **Three corrections are folded into this list** — see What changed from the earlier BOM. The most important: the USB-C CC resistors were missing, and without them the unit will not charge from a USB-C charger at all.

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1. On-board electronics

ACTIVE PARTS

REF	PART	PACKAGE	QTY	SOURCE	UNIT COST	NOTES
U1	ESP32-WROOM-32E (not S3/C3)	Module	1	LCSC C701343	~$2.20	Classic ESP32 keeps the GPIO4/GPIO21/GPIO22/GPIO34 mapping the firmware assumes
U2	TP4056 (bare IC)	SOP-8	1	LCSC C382139	~$0.10	Charge controller
U3	DW01A	SOT-23-6	1	LCSC (search DW01A)	~$0.05	Over-discharge/over-current protection — **not** included in a bare TP4056
U4	FS8205A	SOT-23-6	1	LCSC (search FS8205A)	~$0.08	Dual MOSFET pair driven by U3
U5	XC6206P332MR (3.3V LDO)	SOT-23-3	1	LCSC C5446	~$0.03	**Not AMS1117** — quiescent current must stay in single-digit µA or deep sleep is pointless
U6	ADXL345	LGA-14	1	LCSC (search ADXL345BCCZ)	~$1.80	Tilt sensing. I²C mode: SDO→GND (addr 0x53), CS→3V3
U7	**MCP9808**	DFN-8	1	LCSC (search MCP9808T)	~$0.50–1	Temperature. Same I²C bus as U6; A0/A1/A2→GND gives 0x18. **Must be thermally bonded to the housing wall** — see below
Q1	AO3401 PMOS	SOT-23	1	LCSC C15127	~$0.02	Gates the battery divider so it draws nothing in sleep
Q2	MMBT3904 NPN	SOT-23	1	LCSC C8598	~$0.02	Low-side switch for Q1's gate — a GPIO **cannot** drive the PMOS gate directly, see `PCB_DESIGN.md`
D1	LED, red	0603	1	—	~$0.02	TP4056 CHRG. Optional, but the only charge feedback on a sealed unit
D2	LED, green	0603	1	—	~$0.02	TP4056 STDBY (charge complete)

Subtotal: ~$4.34

RESISTORS (ALL 0603)

REF	VALUE	QTY	PURPOSE
R1, R2	100kΩ **1%**	2	Battery divider. 1% matters — the firmware assumes an exact 1:1 ratio
R3	4.7kΩ	1	DS18B20 1-Wire pull-up. **Only needed if you fit a DS18B20 instead of the MCP9808** — omit otherwise, along with the GPIO4 net
R4	10kΩ	1	EN pull-up
R5	10kΩ	1	IO0 pull-up
R6	100kΩ	1	Q1 gate pull-up to BAT+ — this, not the GPIO, defines the solidly-off state
R7, R8	4.7kΩ	2	I²C pull-ups (SDA/SCL to the ADXL345)
R9	1.2kΩ	1	TP4056 PROG — ~1A, i.e. 0.40C for a 2500mAh 18650. **Re-size if you change cells**, see below
R10	10kΩ	1	Q2 base resistor
R11, R12	5.1kΩ **1%**	2	**USB-C CC1/CC2 pull-downs — required, see below**
R13, R14	2kΩ	2	D1/D2 LED current limit

14 resistors, ~$0.14 total

CAPACITORS

REF	VALUE	PACKAGE	QTY	PURPOSE
C1	1µF	0603	1	EN RC delay (with R4)
C2, C3	0.1µF	0603	2	ESP32 module VDD decoupling
C4	22µF	0805	1	3V3 bulk. WiFi TX pulls 250–500mA in bursts that an LDO alone won't follow
C5	10µF	0805	1	TP4056 VIN
C6	10µF	0805	1	TP4056 BAT
C7	1µF	0603	1	LDO input
C8	1µF	0603	1	LDO output
C9	0.1µF	0603	1	Battery-ADC node filter (the 30ms settling delay in firmware is sized for this cap)
C10	0.1µF	0603	1	ADXL345 VDD I/O decoupling
C11	1µF	0603	1	ADXL345 VS — the datasheet asks for 1µF here, not 0.1µF

11 capacitors, ~$0.28 total

CONNECTORS AND MECHANICAL (BOARD-MOUNTED)

REF	PART	QTY	UNIT COST	NOTES
J1	USB-C 16-pin receptacle, power-only	1	~$0.15	LCSC C165948. **Mounts on the top cap, above the waterline**
J3	JST-PH 2-pin	1	~$0.08	Cell leads. Omit and solder directly if the housing is tight
J4	2.54mm 6-pin header	1	~$0.05	Assembly-time programming only — not populated in the field
SW1	Tactile button	1	~$0.03	IO0 boot-mode select. Optional if you program via J4

No J2. The Hub has a JST probe connector; this board deliberately has none. The whole unit is submerged, and an external probe would mean a cable penetration through the housing — one more seal to fail.

On-board parts subtotal: ~$5.07. In practice you don't buy these individually: budget **$6–8/unit (R165–R220)** for PCB fab + SMT assembly with all of the above populated, at 15–25 unit quantity.

2. Off-board parts

PART	QTY	UNIT COST	NOTES
DS18B20, bare TO-92 — *fallback only*	0–1	$1–2	Superseded by the on-board MCP9808 (U7). Keep this option if you'd rather glue a discrete sensor exactly where you want it than thermally bond the PCB to the wall. Not the cabled probe — that's for the bench rig
18650 Li-ion cell (~2500mAh) + holder or solder tabs	1	$2–4	Fits the 37mm housing with room to spare. R9 is sized for this; a different capacity means a different R9
Sealed floating housing, **~37mm × 125mm** (tube + top cap)	1	$3–7	The real cost and schedule risk. Sized against the commercial precedent — see below
Ballast — lead-free shot, stainless nuts, or tungsten putty	~20–50g	$0.30–1.50	**Required.** Sets the float angle; without it the unit floats flat and reads nothing useful. A bigger body displaces more, so it needs more ballast
Thermal epoxy / potting compound	~2g	$0.50–1	Bonds the DS18B20 to the housing wall. Thermally conductive, electrically insulating
O-ring or silicone gasket for the cap	1	$0.10–0.30	Size to the housing
Desiccant sachet (silica gel)	1	$0.05–0.15	Stops internal condensation as the unit cycles between fermenter and room temperature
Closed-cell foam spacer / double-sided tape	1	$0.10–0.20	Stops the pack shifting, which would change the float angle mid-batch
USB-C port plug	1	$0.10–0.30	Keeps wort out of the connector while submerged

Off-board subtotal: ~$7.15–16.45. Optional: silicone conformal coating on the assembled board (~$0.20/unit amortized) — cheap insurance in a device whose failure mode is water ingress.

3. Totals

	USD	ZAR (LANDED)
PCB fab + SMT assembly (all on-board parts)	$6–8	R165–R220
Off-board parts	$7.15–16.45	R195–R450
Per unit	**~$13–25**	**~R360–R690**

For a 15–25 unit batch: roughly **R5,400–R17,250**. Order and leak-test housings *before* committing to a PCB run — the board is proven architecture reused from the Hub, the housing is not.

What changed from the earlier BOM

The table in `MULTI_SENSOR_MODULES.md` and the schematic disagreed with each other. Reconciling them for this list turned up three things:

1. The USB-C CC resistors were missing entirely — this one is functional. A USB-C receptacle acting as a *sink* must present 5.1kΩ from CC1 and CC2 to ground. A USB-C charger detects those pull-downs before it enables VBUS; with no Rd present it supplies nothing and the unit never charges. The trap is that it appears to work on the bench, because a legacy USB-A→C cable carries its own pull-up in the cable assembly and delivers VBUS regardless. So the fault shows up only when a customer uses a modern C-to-C charger. The Hub's `PCB_DESIGN.md` had it right in the DFM checklist but called the resistors "optional" in the BOM line for J1, and neither BOM listed them as parts. Added here as R11/R12.

2. R7 and R8 were used twice in the schematic — once as the 5.1k CC pull-downs and again as the 4.7k I²C pull-ups. Two different parts sharing a reference designator produces a broken netlist and a wrong pick-and-place file. Renumbered: R7/R8 are the I²C pull-ups, R11/R12 are the CC resistors.

3. The capacitor list didn't match the schematic. The BOM called C5/C6 the TP4056 input/output caps, but the schematic drew them as the LDO's input and output caps, leaving the TP4056 with none and no bulk cap on 3V3. Both sets are needed, so they're separated here (C5/C6 for the TP4056, C7/C8 for the LDO, C4 for 3V3 bulk), and ADXL345 decoupling is split into C10/C11 to match its datasheet.

Also added, because none of it was previously listed and you cannot build a working floating hydrometer without it: **ballast, potting compound, cap gasket, desiccant, foam spacer, and a port plug.**

Before you order

R9 is set for a 2500mAh 18650. The TP4056's charge current is `I_charge ≈ 1200 / R_prog` (amps, ohms), so **1.2kΩ gives ~1A** — a comfortable 0.40C for a 2500mAh cell, and the same value the Hub uses.

Charge current	~1A (0.40C on 2500mAh)
Charge time	~2.5h constant-current, ~3–4h including the CV tail
Termination	~100mA (the TP4056 stops at ~1/10 of the programmed current)

If you change cells, change R9. A smaller cell needs a smaller current or you exceed its safe charge rate — 1A into a 350mAh 10440 would be ~2.9C, roughly triple its rating, which is a cell-damage and venting risk rather than merely poor cycle life. Points on the curve, at a conservative 0.5C:

CELL CAPACITY	0.5C TARGET	R9 (E24)	ACTUAL
350mAh (10440)	175mA	6.8kΩ	176mA
500mAh	250mA	4.7kΩ	255mA
800mAh	400mA	3.0kΩ	400mA
2000mAh (18650)	1000mA	1.2kΩ	1000mA (0.50C)
2500mAh (18650)	—	**1.2kΩ**	**1000mA (0.40C)**
3400mAh (18650)	—	1.2kΩ	1000mA (0.29C)

Temperature sensor: on the board, not on a probe — but coupling decides it. Because the whole unit is submerged, the sensor never needed to be a cabled probe. The board carries an **MCP9808** on the ADXL345's existing I²C bus: ±0.25 °C against the DS18B20's ±0.5 °C, cheaper, no extra GPIO, no 1-Wire pull-up, and no exposure to the counterfeit DS18B20 problem. (For reference, KegLand's RAPT Pill claims ±0.2 °C.)

Placement dominates the datasheet by an order of magnitude. A ±0.25 °C part reading the enclosure's still internal air is far worse than a ±0.5 °C part bonded to the housing wall. So the board must sit against the wall with thermal gap filler at the sensor, and the layout needs copper relief keeping U7 away from the ESP32 and the LDO. If your housing makes that impractical, fit a discrete DS18B20 on two short wires and glue it to the wall instead — the firmware auto-detects which sensor is present and runs either way.

For gravity specifically the sensor grade barely matters: sensitivity is ~0.21 gravity points per °C, so even ±0.5 °C costs only ±0.11 points against a ±2 point target. The case for the MCP9808 is the customer-facing temperature spec, not the compensation.

Why 37mm, and why that matters more than it sounds. An earlier revision of this BOM assumed a 25–32mm test-tube-style housing, which forced a slim 10440 or pouch cell of ~350mAh and gave an estimated **~3 weeks** of battery life at the firmware's 15-minute interval. The commercial precedent is wider: KegLand's RAPT Pill is 37 × 125mm and runs an 18650. At 37mm an 18650 (18mm diameter) fits easily, and ~2500mAh takes the same duty cycle to an estimated **~5 months** — matching the "months between charges" a shipping product claims, which is a useful sanity check on the estimate.

That single dimension change is the difference between charging between every batch and charging twice a year. It also leaves room for a ~33mm Qi receiver coil, close to standard wearable-class parts, if the port-free charging route is taken later.

Estimated runtime at 2500mAh usable-2200mAh, by reporting interval:

INTERVAL	PER DAY	RUNTIME
10 min	20.5mAh	~3.6 months
15 min (firmware default)	**13.8mAh**	**~5.3 months**
30 min	7.1mAh	~10 months

These are estimates from the firmware's `SLEEP_INTERVAL_SECONDS = 900` and typical ESP32 WiFi draw, not measurements. A marginal WiFi link makes them much worse — a failed connect burns the full 15-second timeout instead of ~5 seconds, roughly tripling wake energy.

The housing still decides several other parts. Diameter sets which cell fits; length and total mass set how much ballast you need; the cap sets the gasket. Don't order cells or ballast before you have a housing in hand.

Two things remain unvalidated on hardware, both flagged in `MULTI_SENSOR_MODULES.md`:

- **The tilt axis assumption** — which physical accelerometer axis runs along the housing. It's step 1 of the calibration procedure in `firmware/README.md` precisely because it's a guess until measured.
- **Housing waterproofing.** A leak destroys the unit *and* the batch it's floating in. Leak-test a housing, weighted and sealed, in water at fermentation temperature for several days before trusting one to a batch.

None of this has been built or tested physically — it is a design on paper, reusing a power section that is itself validated only on the Hub's schematic.